

# Understanding User

## Common Problems with Usability

Usability assessment should begin in the early stages of the product development cycle and should be continually applied throughout the process. The assessment should include the user's entire experience, and all the product's important components.

### 10 Most Common Usability Problems:

1. Ambiguous menus and icons.
2. Languages that permit only single-direction movement through a system.
3. Input and direct manipulation limits.
4. Highlighting and selection limitations.
5. Unclear step sequences.
6. More steps to manage the interface than to perform tasks.
7. Complex linkage between and within applications.
8. Inadequate feedback and confirmation.
9. Lack of system anticipation and intelligence.
10. Inadequate error messages, help, tutorials, and documentation.

The Web, with its dynamic capabilities and explosive entrance into our lives, has unleashed what seems like more than its own share of usability problems. Many are similar to those outlined above. Web usability characteristics particularly wasteful of people's time, and often quite irritating, are:

- **Visual clutter:** Excessive decorations, unnecessary graphics, and lack of white space create visual noise, making it difficult for users to find relevant content and causing them to waste time searching.
- **Impaired readability:** Poor choices in fonts, colors, and backgrounds reduce readability, drawing attention away from content and forcing users to focus on deciphering the design rather than understanding the information.
- **Incomprehensible components:** Some design elements, like unclear icons or buttons that don't appear clickable, confuse users by not indicating their function, making it hard to interact with the page.
- **Annoying distractions:** Constantly moving elements, blinking text, and automatic sounds distract users, interrupting their concentration and reducing the page's overall readability.
- **Confusing navigation:** Poor site structure and unclear navigation can leave users feeling lost, with unexpected paths, dead-ends, and elements that don't guide them effectively through the content.

## Human Characteristics and Consideration in Design

### KNOWLEDGE/EXPERIENCE

|                            |                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------|
| Computer Literacy          | Highly technical or experienced, moderate computer experience, or none.                           |
| System Experience          | High, moderate, or low knowledge of a particular system and its methods of interaction.           |
| Application Experience     | High, moderate, or low knowledge of similar systems.                                              |
| Task Experience            | Level of knowledge of job and job tasks.                                                          |
| Other Systems Use          | Frequent or infrequent use of other systems in doing job.                                         |
| Education                  | High school, college, or advanced degree.                                                         |
| Reading Level              | Less than 5th grade, 5th–12th, more than 12th grade.                                              |
| Typing Skill               | Expert (135 WPM), skilled (90 WPM), good (55 WPM), average (40 WPM), or "hunt and peck" (10 WPM). |
| Native Language or Culture | English, another, or several.                                                                     |

### JOB/TASK/NEED

|                         |                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------|
| Type of System Use      | Mandatory or discretionary use of the system.                                             |
| Frequency of Use        | Continual, frequent, occasional, or once-in-a-lifetime use of system.                     |
| Task or Need Importance | High, moderate, or low importance of the task being performed.                            |
| Task Structure          | Repetitiveness or predictability of tasks being automated, high, moderate, or low.        |
| Social Interactions     | Verbal communication with another person required or not required.                        |
| Primary Training        | Extensive or formal training, self-training through manuals, or no training.              |
| Turnover Rate           | High, moderate, or low turnover rate for jobholders.                                      |
| Job Category            | Executive, manager, professional, secretary, clerk.                                       |
| Lifestyle               | For Web e-commerce systems, includes hobbies, recreational pursuits, and economic status. |

### PSYCHOLOGICAL CHARACTERISTICS

|                 |                                                                     |
|-----------------|---------------------------------------------------------------------|
| Attitude        | Positive, neutral, or negative feeling toward job or system.        |
| Motivation      | Low, moderate, or high due to interest or fear.                     |
| Patience        | Patience or impatience expected in accomplishing goal.              |
| Expectations    | Kinds and reasonableness.                                           |
| Stress Level    | High, some, or no stress generally resulting from task performance. |
| Cognitive Style | Verbal or spatial, analytic or intuitive, concrete or abstract.     |

### PHYSICAL CHARACTERISTICS

|              |                                                    |
|--------------|----------------------------------------------------|
| Age          | Young, middle aged, or elderly.                    |
| Gender       | Male or female.                                    |
| Handedness   | Left, right, or ambidextrous.                      |
| Disabilities | Blind, defective vision, deafness, motor handicap. |

## **Human Considerations in Screen Design**

Use of a screen, and a system, is affected by many factors. These include: how much information is presented on a screen, how a screen is organized, the language used on the screen, the distinctiveness of the screen's components, its aesthetics, and a screen's consistency with other screens. First, let's look at what aspects of poor screen design can be distracting to the user, what a user is looking for in good design, and the kinds of things screen do interactions with a system or a web site.

### **What Screen Users Want?**

What are people looking for in the design of screens? One organization asked a group of screen users and got the following responses:

- An orderly, clean, clutter-free appearance.
- An obvious indication of what is being shown and what should be done with it.
- Expected information located where it should be.
- A clear indication of what relates to what, including options, headings, captions, data, and so forth.
- Plain, simple English.
- A simple way of finding out what is in a system and how to get it out.
- A clear indication of when an action can make a permanent change in the data or system.
- The desired direction is toward simplicity, clarity, and understandability—qualities lacking in many of today's screens.

### **What Screen Users do?**

- Task identification: Users identify a task or need, ranging from highly structured activities like data entry to more open-ended inquiries, such as finding the best local rehabilitation program or planning a vacation.
- Task completion decision: Users decide how to complete the task by selecting the appropriate transaction screens or browsing through relevant screens for more general needs.
- Control manipulation: Users interact with the computer's controls (keyboard, mouse, etc.) to make selections, input data, and execute commands.
- Data gathering: Users collect necessary information from sources like forms or coworkers, either by directly inputting it or by navigating through multiple screens or web links.
- Decision-making: Users make judgments based on the gathered data, determining if the task is complete, whether additional information is needed, or which steps to take next.

## **Interface Design Goals**

To make an interface easy and pleasant to use, then, the goal in design is to:

- Reduce visual work.
- Reduce intellectual work.
- Reduce memory work.
- Reduce motor work.
- Minimize or eliminate any burdens or instructions imposed by technology.

The result will always be improved user productivity and increased satisfaction.

## **Responses to Poor Design**

- Psychological:
  - Panic or stress - Over complex systems and procedures, unexpected long delays during times of severe or unusual pressure, or long response times.
  - Boredom - Boredom results from, among other things, improper computer pacing (slow response times or long download times) or over simplistic jobs or tasks. A bored individual is also likely to make more performance errors.
- Physical: When people do something, they expect the benefits of what they are doing to outweigh the cost or effort to do it. The following physical reactions are:
  - Abandonment of the system - In business systems this was a common reaction of managerial and professional personnel.  
With the Web, almost all users can exercise this option.  
Partial use of the system -This is the most common user reaction to most computer systems. Many aspects of many systems often go unused.

Information Collection Techniques:

- Direct methods consist of face-to-face meetings with, or actual viewing of, users to solicit requirements.
  - Individual Face-to-Face Interview
  - Telephone Interview or Survey
  - Traditional Focus Group
  - Facilitated Team Workshop
  - Observational Field Study
  - Requirements Prototyping
  - User-Interface Prototyping
  - Card Sorting for Websites
- Indirect methods impose an intermediary, someone or something, between the users and the developers.
  - MIS Intermediary
  - Paper Survey or Questionnaire
  - Electronic Survey or Questionnaire

- Electronic Focus Group
- Marketing and Sales
- Support Line
- E-Mail, Bulletin Boards, or Guest Book
- User Group
- Competitor Analysis
- Trade Show
- Other Media Analysis
- System Testing

### **Possible Problems in Requirements Collection**

- Not enough user, customer, and other interested party involvement in the process. The result may be incomplete requirements.
- Lack of requirements management or coordination. Requirement modifications are not recorded, tracked properly, or carried out. Requirements may be inaccurate or incomplete.
- Communication problems among all participants. It is possible that not all participants understand the exact requirements.
- Capturing the relevant information may be difficult. The relevant knowledge may exist in many places and locations, including books, operating manuals, and in people's heads.
- People who do understand the problem may be constrained. Heavy workloads and a lack of time may make people reluctant to, or unable to, participate.
- Organizational and political factors and agendas may influence the process. The resulting views may not tally with the users views. Resistance to new ideas and change may be reflected in comments made.
- Disparities in knowledge may exist. Some people may know what they want only in general terms, whereas others may be forceful and detailed about their wants. Getting a balanced view may be difficult.
- Changing economic and business environments and personnel roles. As a result, want and needs change as well.

## Ordering of Screen Data and Content

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- Divide information into units that are logical, meaningful, and sensible.
  - Organize by the degree interrelationship between data or information.
  - Provide an ordering of screen units of information and elements that is prioritized according to the user's expectations and needs.
  - Possible ordering schemes include:
    - Conventional.
    - Sequence of use.
    - Frequency of use.
    - Function.
    - Importance.
    - General to specific.
  - Form groups that cover all possibilities.
  - Ensure that information that must be compared is visible at the same time.
  - Ensure that only information relative to the users tasks or needs is presented on the screen.
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### Upper Left Starting Point:

Provide an obvious starting point in the screen's upper-left corner.

## Screen Navigation and Flow

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- Provide an ordering of screen information and elements that:
    - Is rhythmic, guiding a person's eye through the display.
    - Encourages natural movement sequences.
    - Minimizes pointer and eye movement distances.
  - Locate the most important and most frequently used elements or controls at the top left.
  - Maintain a top-to-bottom, left-to-right flow.
  - Assist in navigation through a screen by:
    - Aligning elements.
    - Grouping elements.
    - Using of line borders.
  - Through focus and emphasis, sequentially, direct attention to items that are:
    1. Critical.
    2. Important.
    3. Secondary.
    4. Peripheral.
  - Tab through window in logical order of displayed information.
  - Locate command buttons at end of the tabbing order sequence.
  - When groups of related information must be broken and displayed on separate screens, provide breaks at logical or natural points in the information flow.
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## **Visually Pleasing Composition:**

- Balance: Create screen balance by providing an equal weight of screen elements, left and right, top and bottom.
- Symmetry: Create symmetry by replicating elements left and right of the screen centerline.
- Regularity: Create regularity by establishing standard and consistently spaced horizontal and vertical alignment points. Also, use similar element sizes, shapes, colors, and spacing.
- Predictability: Create predictability by being consistent and following conventional orders or arrangements.
- Sequentiality: Provide sequentiality by arranging elements to guide the eye through the screen in an obvious, logical, rhythmic, and efficient manner. The eye tends to be attracted to:
  - A brighter element before one less bright.
  - Isolated elements before elements in a group.
  - Graphics before text.
  - Color before black and white.
  - Highly saturated colors before those less saturated.
  - Dark areas before light areas.
  - A big element before a small one.
  - An unusual shape before a usual one.
  - Big objects before little objects.
- Economy: Provide economy by using as few styles, display techniques, and colors as possible.
- Unity: Create unity by:
  - Using similar sizes, shapes, or colors for related information.
  - Leaving less space between elements of a screen than the space left at the margins.
- Proportion: Create windows and groupings of data or text with aesthetically pleasing proportions.
- Simplicity: Optimize the number of elements on a screen, within limits of clarity. Minimize the alignment points, especially horizontal or columnar.
  - Provide standard grids of horizontal and vertical lines to position elements.
- Groupings: Provide functional groupings of associated elements. Create spatial groupings as closely as possible to five degrees of visual angle. Visually reinforce groupings:
  - Provide adequate separation between groupings through liberal use of white space.
  - Provide line borders around groups.
  - Provide meaningful titles for each grouping.

## ***Visual Style in Web Page Design***

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- Maintain a consistent and unified visual style throughout the pages of an entire Web site.
  - Base the visual style on:
    - The profile and goals of the Web site owner.
    - The profile, tastes, and expectations of the Web site user.
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## **Amount of Information**

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- Present the proper amount of information for the task.
    - Too little is inefficient.
    - Too much is confusing.
  - Present all information necessary for performing an action or making a decision on one screen, whenever possible.
    - People should not have to remember things from one screen to the next.
  - Restrict screen or window density levels to no more than about 30 percent.
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### **Design Principles with Composition**

- Composition is how we see information. It influences the relationships we see between objects, perceptions of complexity, and affects our desire to engage with content.
- A balanced composition feels right. It feels stable and aesthetically pleasing. While some of its elements might be focal points and attract your eye, no one area of the composition draws your eye so much that you can't see the other areas.

### **Gestalt' Principles:**

- Emergence: People tend to identify elements first in their general outlined form. Our brain recognises a simple, well- defined object quicker than a detailed one.
- Reification: People can recognise objects even when there are parts of them missing. Our brain matches what we see with familiar patterns stored in our memory and fills in the gaps.
- Invariance: People can recognise simple objects independently of their rotation, scale and translation. Our brain can perceive objects from different perspectives, despite their different appearance.
- Multi-Stability: People will often interpret ambiguous objects in more than one way. Our brains will bounce back and forth between the alternatives seeking certainty. As a result, one view will become more dominant while the other one will get harder to see.
- Proximity: Elements arranged close to each other are perceived as more related than those placed further apart. This way different elements are viewed mainly as a group rather than as individual elements.